MIL-D-50082D
30 SEPTEMBER 1964
SUPERSEDING
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MILITARY SPECIFICATION

DISCS, INLET AND NOSECUP VALVE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers four types of natural rubber valve discs.

1.2 Classification. Valve discs shall be of the following types, as specified (see 6.2):

Type I — Inlet valve disc (Dwg A5–2–1004).

Type II — Nosecup valve disc (Dwg A5–2–1005).

Type III — Inlet valve disc (Dwg B5–77–940–2).

Type IV — Nosecup valve disc (Dwg B5–77–940–1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-601 — Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

A5–2–1004 — Disc, Valve.

A5–2–1005 — Disc, Valve.

B5–2–464 — Valve, Nosecap.

C5–2–980 — Cover, Cap-Inlet Valve.

B5–77–940 — Disc, Valve.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

FSC 4240

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3. REQUIREMENTS

3.1 Material.

3.1.1 *Rubber compound.* The rubber compound used in the manufacture of the discs shall contain no less than 92 percent by volume of new natural rubber.

3.1.2 *Antioxidants and accelerators.* The supplier shall furnish a certificate stating the trade names, the manufacturers, and the amount of the antioxidants, and the accelerators used in compounding the stock.

3.2 Manufacture. The discs shall be die cut from molded sheet stock. Each type as specified in the contract or order (see 6.2) shall be manufactured in accordance with the applicable drawing listed in 2.1.

3.3 Test slabs. Test slabs, 6 by 6 by 0.075 ±0.01 inch thick, shall be made from stock representative of the formulation used for each lot during regular production of discs. The slabs shall have a cure equivalent to that of discs produced during regular production (see 6.5).

3.4 Physical properties. Physical properties of the sheet stock shall conform to the requirements shown in table I when tested as specified therein.

3.5 Distortion. The valve disc shall be free from distortion when tested as specified in 4.6.3.

3.6 Airflow resistance. When assembled to their respective valve seats, the discs shall have a resistance not to exceed a pressure differential of 5 millimeters of water at an airflow rate of 42.5 liters per minute when tested as specified in 4.6.4. The inlet valve discs shall be tested on a cover without a flocked disc.

TABLE I. *Physical properties*

Property	Unaged	After heat aging		Test paragraph
		Types I and II	Types III and IV	
Average tensile strength, psi, min	2600			4.6.2.1
Percent increase, max		20	20	
Percent decrease, max		15	20	
Average ultimate elongation, %, min	600			4.6.2.1
Percent increase, max		20	20	
Percent decrease, max		15	20	
Average tension set at 600% elongation, % max	15	.....	.....	4.6.2.1
Average hardness (Shore "A" or "A2")	40 ± 2	.....	.....	4.6.2.1
Flexibility				
Types I and II				
Units at 75° ± 5°F., min	8	.....	.....	
Percent retention at —25 ± 5° F., min	80	.....	.....	
Types III and IV				
Units at 75° ±5°F., min	8	.....	.....	
Percent retention at —50° ± 5°F., min	85	.....	.....	

3.7 Leakage. When assembled to their respective valve seats, the discs shall not leak more than 150 milliliters of air per minute at a pressure differential of 25 millimeters of water when tested as specified in 4.6.5.

3.8 Workmanship. The discs shall have a smooth finish on both sides; shall have clean cut peripheral and hole edges; and shall be free from tackiness, imperfections, such as porous areas, cracks, checks, abrasions, and blisters, foreign matter (grease, oil, dirt, or ash), and damage, such as tears, slits, and punctures.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substation.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the discs of one type produced by one manufacturer from one batch (see 6.3) of raw material and and manufactured under essentially the same manufacturing conditions.

4.4 Sampling.

4.4.1 *For examination and nondestructive tests.* Sampling shall be conducted in accordance with MIL–STD–105.

4.4.2 *For destructive tests.* Sixteen test slabs produced in accordance with 3.3 shall be taken for the tests.

4.5 Inspection provisions.

4.5.1 *For examination and nondestructive tests.* Sample discs shall be examined and tested in accordance with the classification of defects and with MIL–STD–105.

4.5.2 *For destructive tests.* The sample test slabs shall be tested in accordance with 4.6.2.1 and 4.6.2.2. The acceptance number is 0.

4.5.3 *Classification of defects.*

4.5.3.1 *Valve discs (see applicable dwg for type).*

Categories	*Defects*	*Acceptance standards*
Critical:	*None..defined*	
Major:	AQL 2.5 percent defective	
101	Distortion	4.6.3
102	Thickness incorrect	
103	Center hole location incorrect	
104	Center hole diameter incorrect	
105	Disc diameter incorrect	
106	Tackiness	
107	Workmanship (set 3.8)	
Major A:	AQL 0.65 percent defective	
150	Airflow resistance	4.6.4
*151	Leakage	4.6.5
Minor:	AQL 4.0 percent defective	
201	Color incorrect	
202	Peripheral and hole edges incorrect	

* For prime procurement only.

4.6 Tests.

4.6.1 *Preparation of test specimens.* Speci-

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mens shall be prepared from the test slabs (4.4.2) as specified in FED-STD-601. Specimens shall not be buffed or stretched before testing. Specimens shall be prepared for each test in the numbers indicated in table II. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F. for a continuous period of 168 ± 1 hours or at a temperature of 212° ± 1.8°F. for a continuous period of 24 ± 1 hours (see 6.4). NOTE: Accelerated aging at 212°F. can be used only with types I and II discs.

TABLE II. *Number of specimens for various tests*

Test	Specimens	
	Unaged	Aged
Tensile strength	10	10
Ultimate elongation	10	10
Tensile stress	10	10
Hardness	6	—
Tension set	6	—
Flexibility	8	—

4.6.2 *Physical properties tests.*

4.6.2.1 *Tensile strength, ultimate elongation, tensile stress, tension set, and hardness.* Tensile strength, ultimate elongation, tensile stress, tension set, and hardness shall be determined as specified in FED-STD-601.

4.6.2.2 *Flexibility.* Flexibility shall be determined using the Q37 Natural Rubber Flexibility Tester. Flexibility shall be read on the grid of the apparatus. Four specimens of each type shall be tested at 75° ± 5°F; four specimens each of types I and II shall be tested at —25° ± 5°F. after being conditioned for 72 hours at that temperature; and four specimens each of types III and IV shall be tested at —50° ± 5°F. after being conditioned for 24 hours at that temperature.

4.6.3 *Distortion.* The valve discs shall be submerged for several seconds in water at ambient temperature and then shall be permitted to float for five minutes. At the end of the test period, the floating discs shall be observed for distortion. The discs shall then be removed from the water, placed on a flat, dry blotter, and after they are dry, observed again for distortion.

4.6.4 *Airflow resistance.* The valve discs shall be tested for airflow resistance with a Q106 Inlet and Outlet Valve Resistance Indicator after they are assembled to their respective valve seats (Dwgs B5-2-464 or C5-2-980).

4.6.5 *Leakage.* The valve discs shall be tested for leakage (see 6.2) with the M4 Outlet Valve Leakage Indicator after they are assembled to their respective valve seats (Dwgs B5-2-464 or C5-2-980).

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.7)

5.1 Packaging, level C. Discs of one type shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Discs of one type packaged as specified in 5.1 shall be packed in such a manner to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified in the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I discs are intended for use in the inlet valve of protective masks; type II discs are intended for use in the nosecup valve; and types III and IV are intended for use in the E44R1 Winterization Kit.

6.2 Ordering data. Procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) Type of valve disc (see 1.2).

(c) Applicable drawing.

(d) Applicability of leakage test (see 4.5.3).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Aging. Aging at 212° ± 1.8°F. for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 1.8°F. for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens.

6.5 Equivalent cure. Slabs and discs of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.

(b) Their average tensile-stress values at any given elongation, other than ultimate elongation, do not vary more than ± 2.5 percent.

(c) Their curing times can be equated in the following equation:

$$\left(\frac{T_1 - T_2}{10}\right)$$

where: $t_2 = t_1 \times 1.6$

t_1 = Time of cure at temperature T_1 (minutes).

t_2 = Time of cure at temperature T_2 (minutes).

T_1 = Temperature of cure for time t_1 (° F.).

T_2 = Temperature of cure for time t_2 (° F.).

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationships in determining equivalent cure.

6.6 Valve seats. Valve seats required for conducting airflow resistance and leakage tests will be furnished by the Government.

6.7 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheet covering the inlet valve disc should be used. (Copies of repair parts packaging data sheets may be obtained from the Director of Engineering and Industrial Services, ATTN: Chief, Design Branch, Edgewood Arsenal, M-21010.)

6.8 International interest. Certain provisions of this specification are the subject to ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
Army—MU
Navy—SH
Air Force—84

Preparing activity:
Army—MU(EA)
Project No. 4240-0074

International interest (see 6.8)

Review activities:
Army—MD
Navy—SH

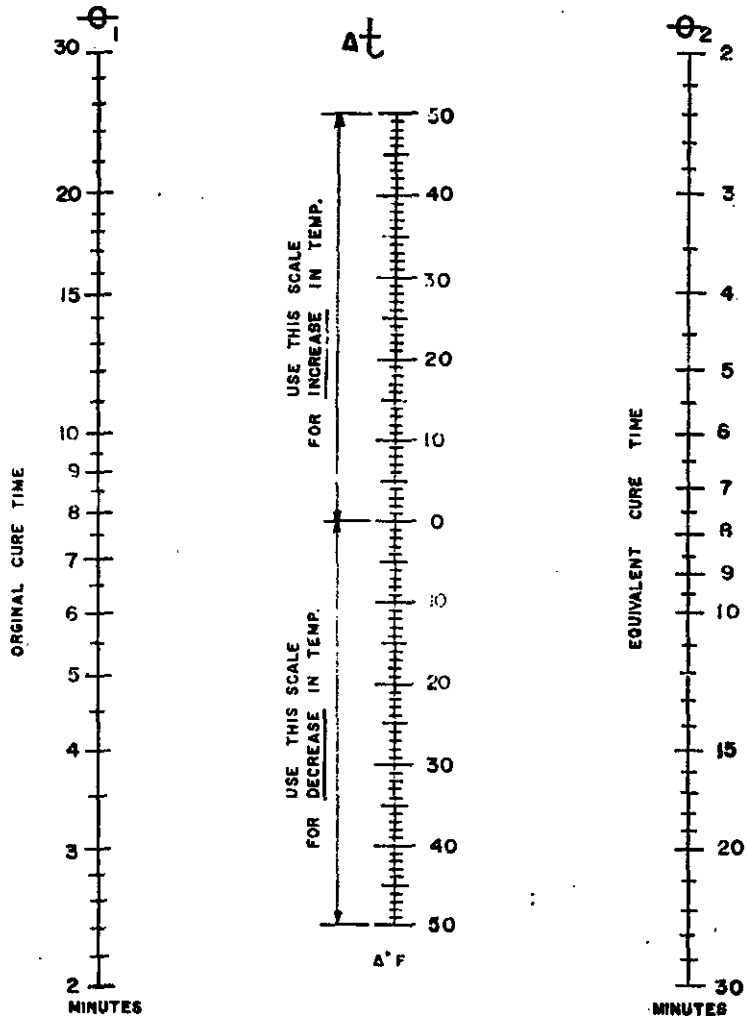
User activities:
Navy—MC
Air Force—84

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EQUIVALENT CURE TIME

FOR

NATURAL RUBBER



USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

FIGURE 1.

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